NESCART-FC REV A-FC – FUNCTIONAL SCHEMATIC

FAMICOM EDGE / MEMORY DATA PATH

BUS CORE

ﾌｧｲﾙ: nescart-fc_bus_core.kicad_sch

UPLOAD / ADDRESS GENERATION

LOADER + INTERFACES

ﾌｧｲﾙ: nescart-fc_loader.kicad_sch

BUS CORE <-> LOADER | all pages share named global nets

POWER / SAFE STATES

POWER + SAFE STATES

ﾌｧｲﾙ: nescart-fc_power_safe.kicad_sch

LOCAL SUPPLY BYPASSING

DECOUPLING

ﾌｧｲﾙ: nescart-fc_decoupling.kicad_sch

Functional flat hierarchy; global nets connect pages

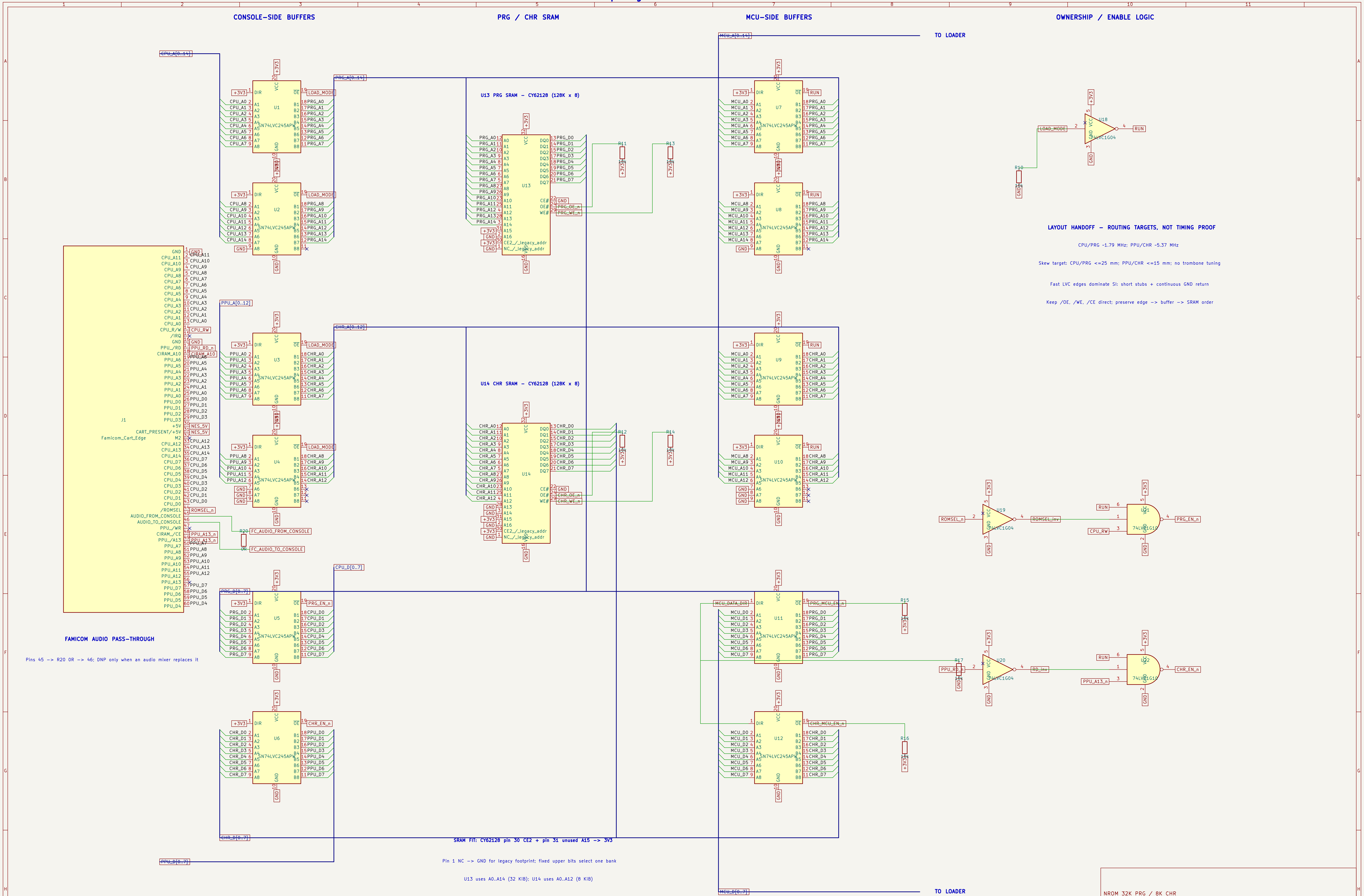
keita

Sheet: /
File: nescart-fc.kicad_sch

Title: nescart-fc Rev A-FC – schematic index

Size: A4	Date:	**Rev: A-FC**
KiCad E.D.A. 10.0.1		Id: 1/5

FAMICOM BUS CORE | edge -> console buffers -> SRAM <- MCU buffers



NROM 32K PRG / 8K CHR

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Sheet: /BUS CORE/

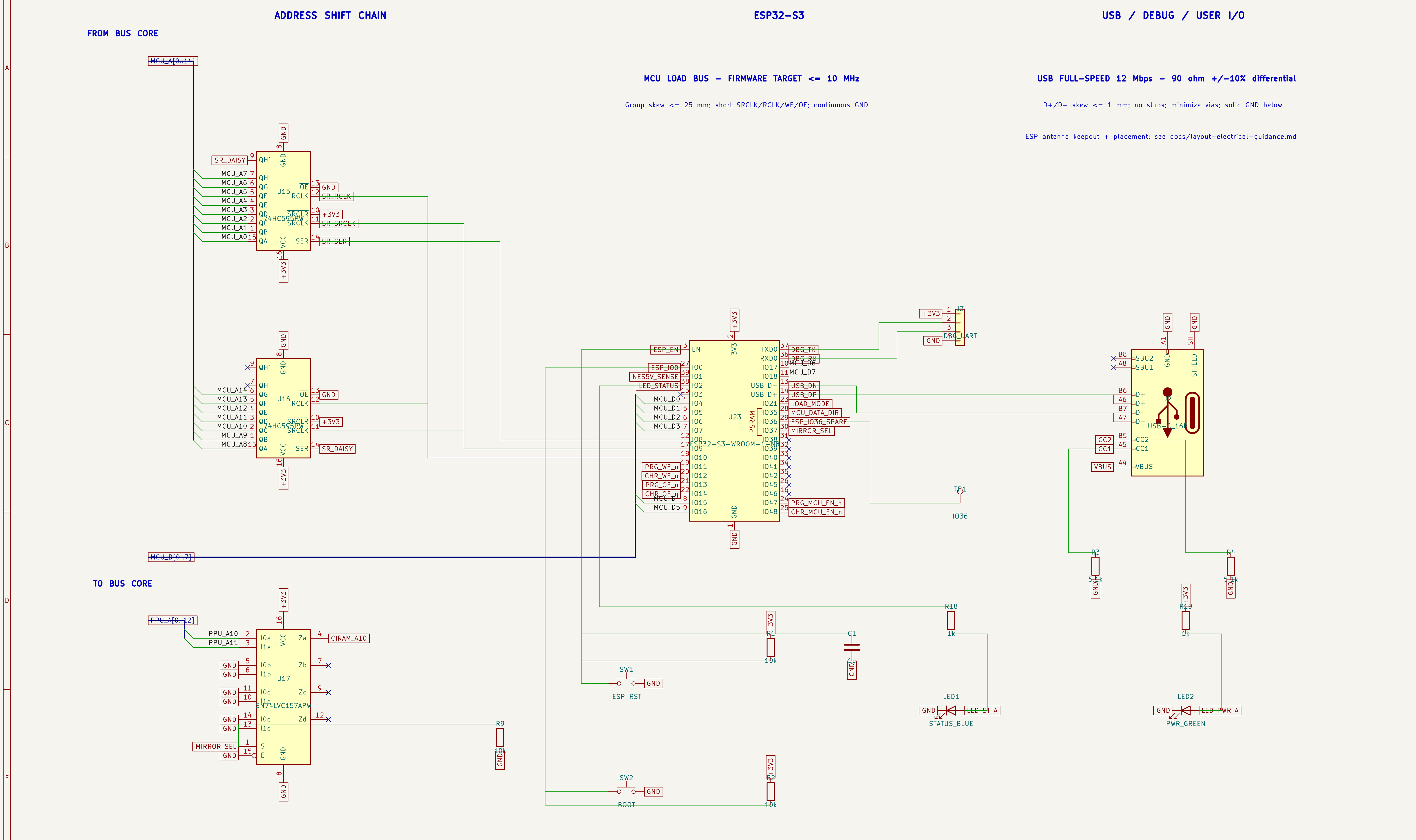
File: nescart-fc_bus_core.kicad_sch

Title: FAMICOM BUS, SRAM, AUDIO, AND OWNERSHIP

Size: A2 | Date: | Rev: A-FC

KiCad E.D.A. 10.0.1 | Id: 2/5

ESP32 LOADER / ADDRESS GENERATION / USB



FAMICOM: no CIC reset hold. Upload completes, then user presses console RESET.

IO36 is spare at TP1; MIRROR_SEL defaults LOW through R9 10k.

NROM 32K PRG / 8K CHR
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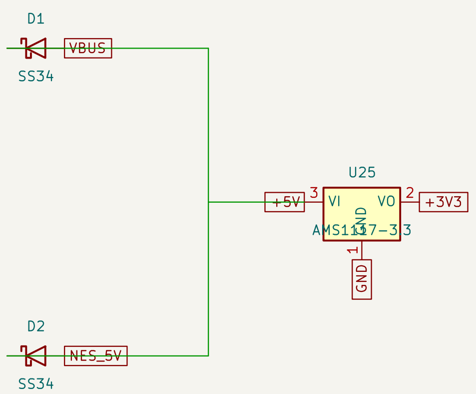
Sheet: /LOADER + INTERFACES/
File: nescart-fc_loader.kicad_sch

Title: ESP32 LOADER AND EXTERNAL INTERFACES

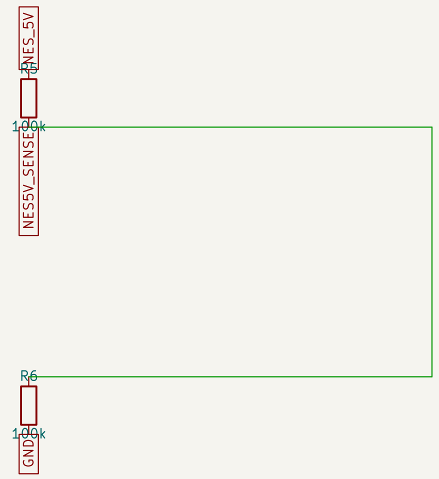
Size: A3 | Date: | Rev: A-FC
KiCad E.D.A. 10.0.1 | Id: 3/5

POWER / STARTUP DEFAULTS

POWER OR + REGULATOR



DEFINED STARTUP STATES



PROVISIONAL POWER-ROUTING ALLOCATION

+3V3 and 5V trunks: design for >= 0.8 A peak

ESP branch supply capacity >= 0.5 A; each SRAM <= 60 mA max

Logic/LED allowance: 0.1 A; measure all rails before fabrication

AMS1117 loss at 0.8 A -= 1.36 W: thermal review REQUIRED

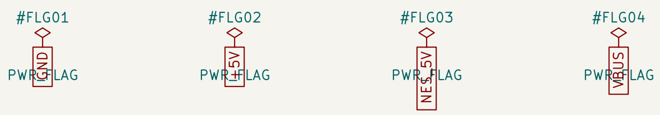
LAYER / RETURN-PATH TARGET

In1: uninterrupted GND plane; In4: +3V3 plane; no signals

Route buses primarily on F.Cu/In2 with In1 GND reference

Keep copper/vias/traces out of ESP antenna zone and Famicom tongue

Source capability + regulator thermals remain signoff items



NROM 32K PRG / 8K CHR
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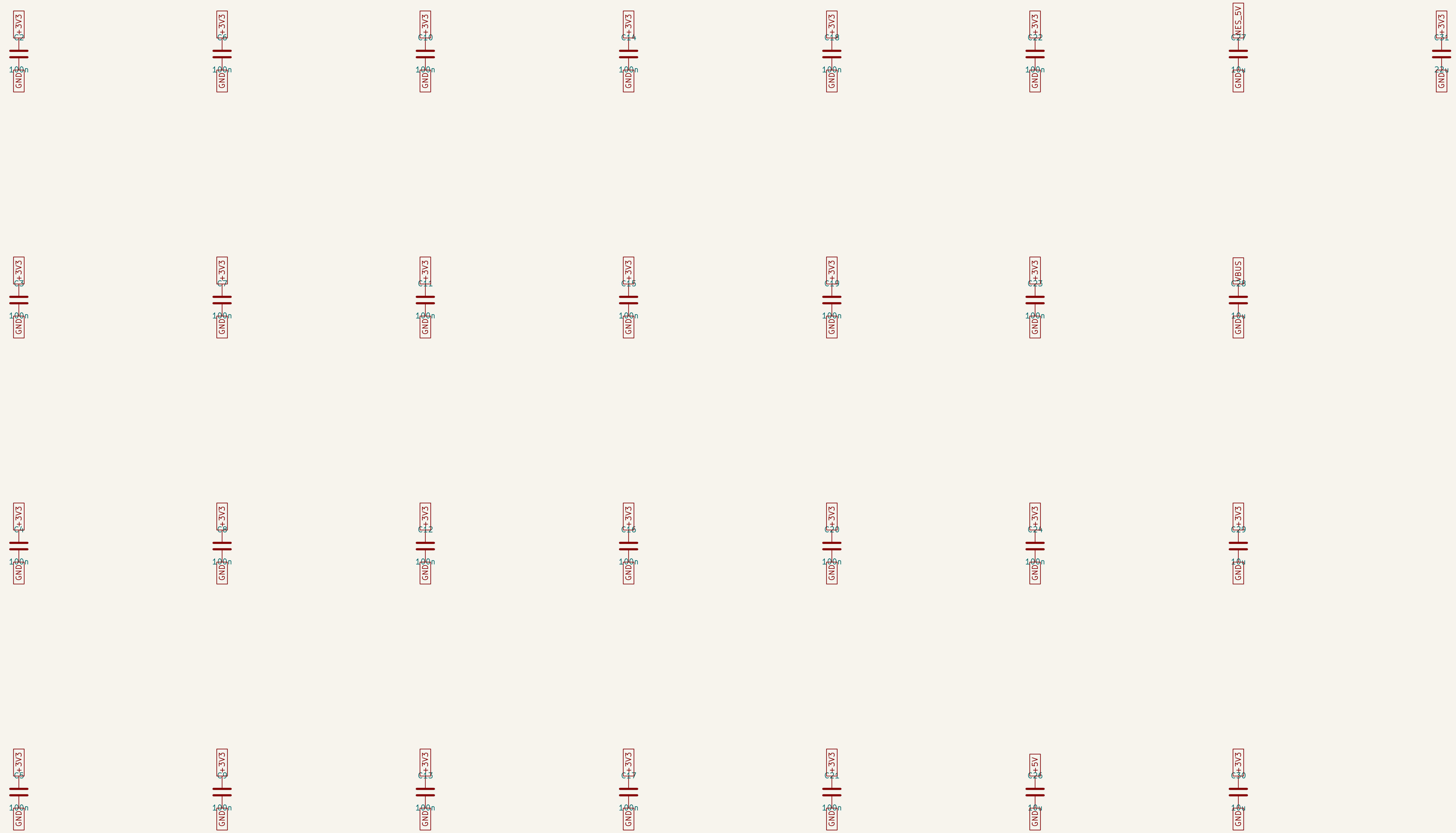
Sheet: /POWER + SAFE STATES/
File: nescart-fc_power_safe.kicad_sch

Title: POWER AND SAFE DEFAULTS

Size: A3	Date:	**Rev: A-FC**
KiCad E.D.A. 10.0.1		Id: 4/5

DECOUPLING / BULK CAPACITORS

One local capacitor per IC plus rail bulk capacitance



LAYOUT: each 100 nF in the VCC-pin escape path with its own short GND via

Place >=10 uF at each source entry and directly at the ESP32 branch

NROM 32K PRG / 8K CHR
keita
Sheet: /DECOUPLING/
File: nescart-fc_decoupling.kicad_sch

Title: DECOUPLING AND BULK CAPACITORS

Size: A3	Date:	**Rev: A-FC**
KiCad E.D.A. 10.0.1		Id: 5/5